Multi-Locus Typing and Genotyping of *Histomonas meleagridis* Isolates

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Histomoniasis

THE PATHOGEN

Histomonas meleagridis

THE TARGET

Devastates turkey flocks

CONSEQUENCES

High mortality rate

Treatment options are not available; therefore, understanding the genetic diversity is an important step toward tracking and studying this parasite

Why Old Tracking Methods Failed

THE OLD METHOD

ITS region to track parasites

FLAWS

Intra-isolate heterogeneity
Clonal cultures yielded multiple, conflicting ITS sequence variants

Overlapping Sequences
Identical sequence variants among completely separate isolates

An identity tag is useless if it changes constantly.
ITS markers showed limitations for reliable subtyping.

THE NEW METHOD

The Multi-Locus Investigation

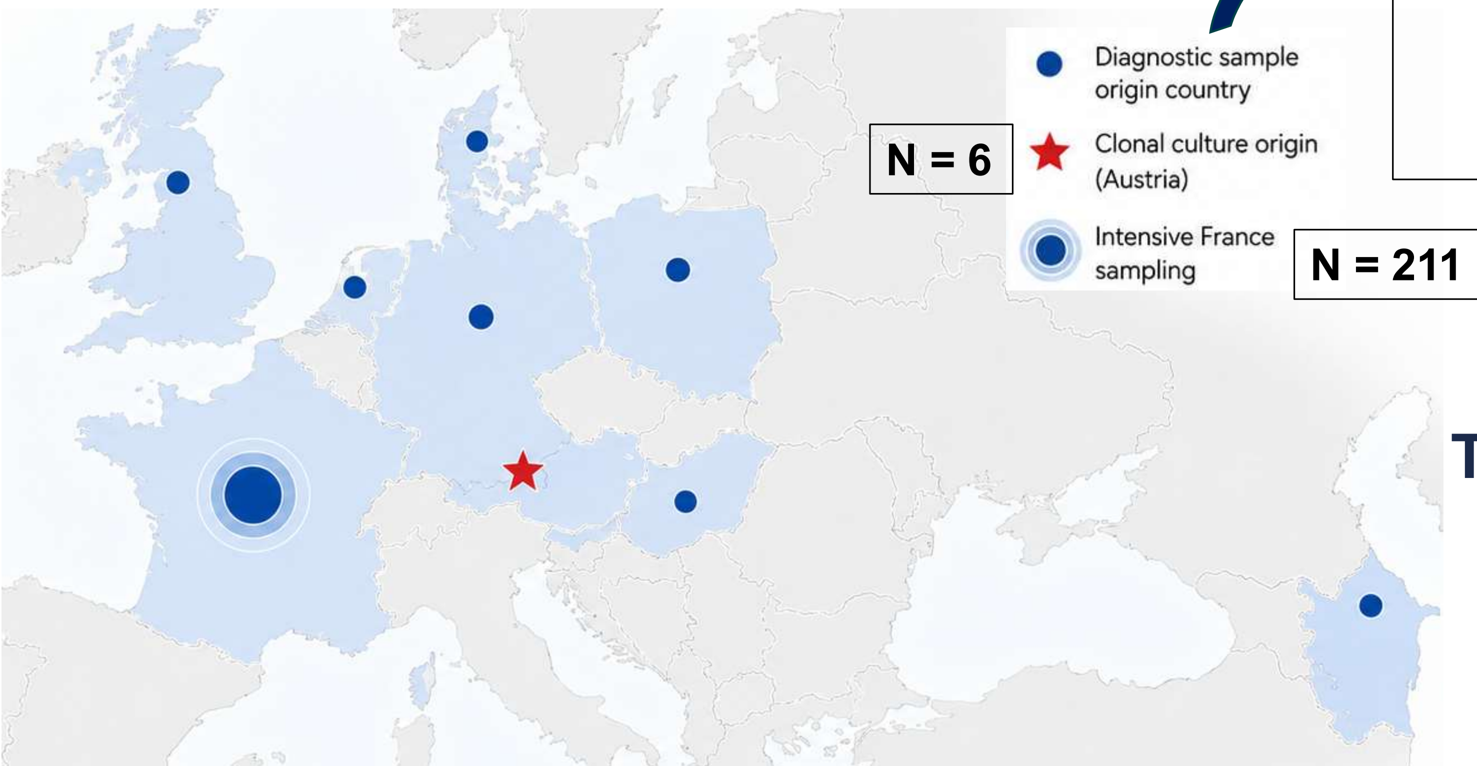
Objectives

To investigate the genetic heterogeneity of *Histomonas meleagridis* isolates using a multi-locus sequencing approach

To identify the genetic markers useful for molecular subtyping of *H. meleagridis*

The Multi-Locus Investigation

Azerbaijan, Denmark, France, Germany, Hungary, the Netherlands, Poland, and the United Kingdom
N = 39



Total samples: 256

Geographic origin of the samples

Workflow

18S rRNA
Targeting broad genetic separation

rpb1
Protein-coding, single-copy candidate

alpha-actinin1
Protein-coding, single-copy candidate

DNA extraction and PCR amplification

Sequencing

Sequence Analysis

Assemble + align sequences

Assembly: Accelrys Gene and Lasergene
Alignment: **ClustalX v2.1**

Confirm identity

BLAST used especially for 18S rRNA products because the 18S assay amplified related protozoa as well.

Build phylogenies

PHYLIP v3.68:
Neighbor-Joining
Maximum likelihood

Test tree stability

Bootstrap
1000 replicates for distance trees
100 replicates for ML trees

18S rRNA

Diagnostic Findings

Diversity Score: High. Yielded 34 total sequence types

Identity Range: 95.6-99.8% identity between sequences.

Strengths

Both NJ and ML analyses showed two major *H. meleagridis* groups

Limitations

- Useful for broad separation
- Ultra-fine typing should be interpreted cautiously

27 H. meleagridis-specific **sequences**

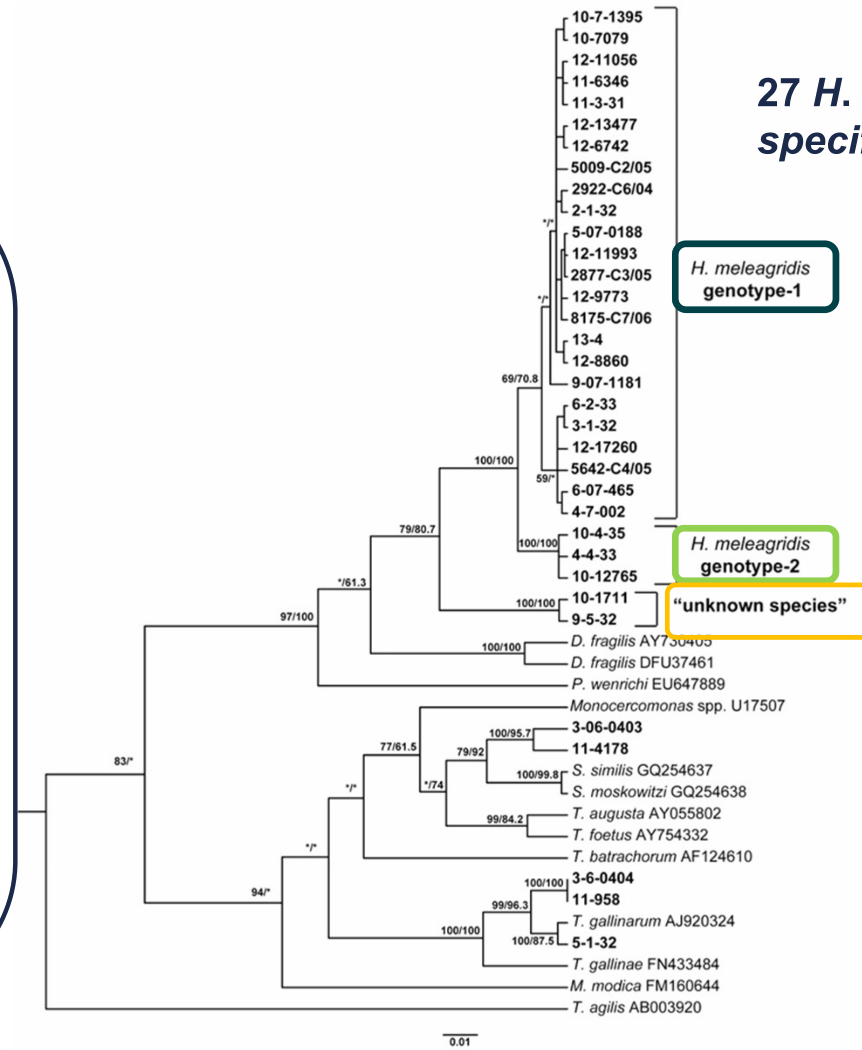


Figure 1. Phylogenetic tree of *H. meleagridis* isolates based on partial 18S rRNA sequences (approx. 544bp).

rbp1

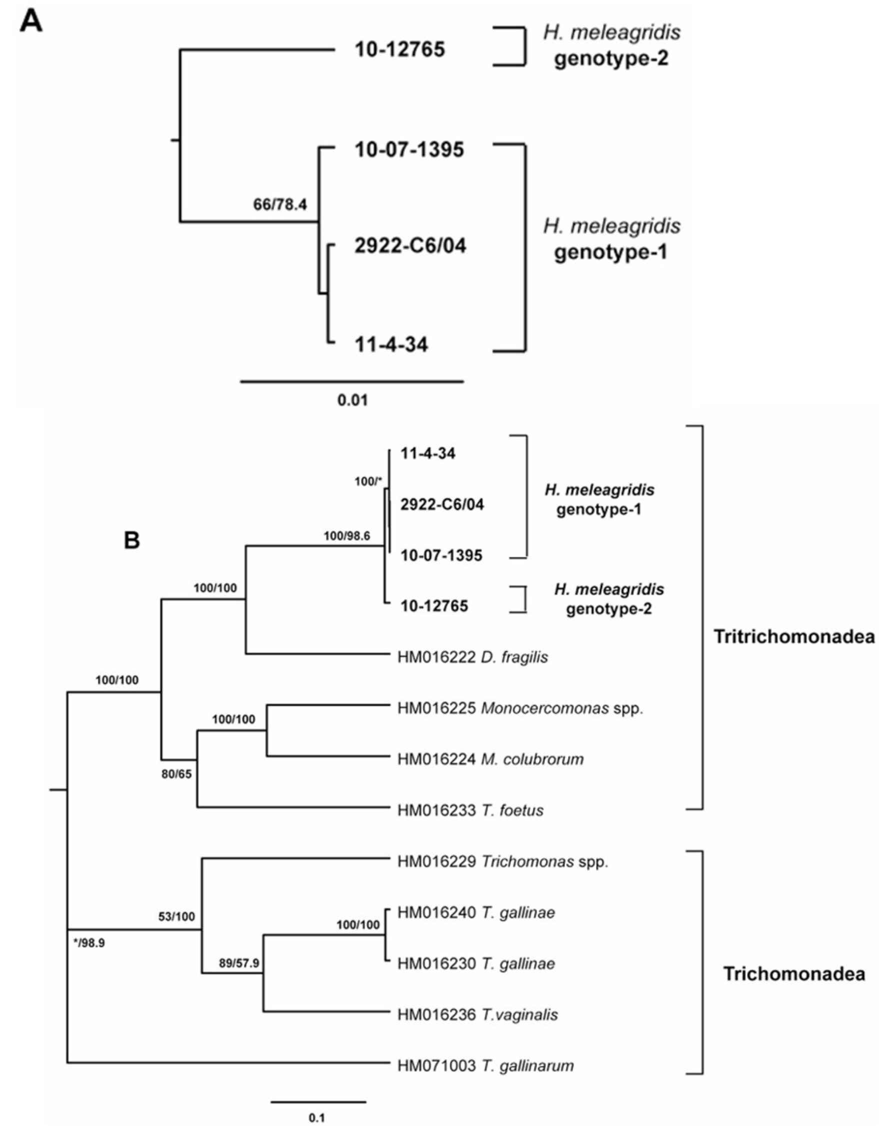
Diagnostic Findings

Diversity Score: Low. Yielded 4 total sequence types.

Differentiation: 17 stable SNPs perfectly separate Genotype 1 and Genotype 2

Adds Resolution

Able to differentiate isolates that the 18S marker identified as identical.



Phylogenetic trees based on partial rpb1 sequences

alpha-actinin1

Diagnostic Findings

Diversity Score: Low. Yielded 3 total sequence types.

Differentiation: 12 stable SNPs between the two clusters, but very high conservation among samples.

Limitations

99.9% identical within Genotype 1.
100% identical within Genotype 2.

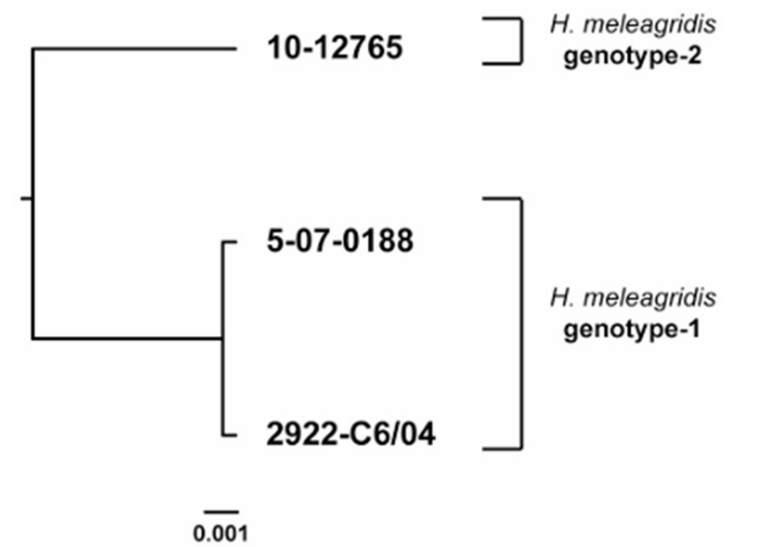


Figure 2. Phylogenetic tree based on partial α-actinin1 sequences of *H. meleagridis* (approx. 1048bp). The phylogenetic

Useful for confirmation, weak for detailed subtyping

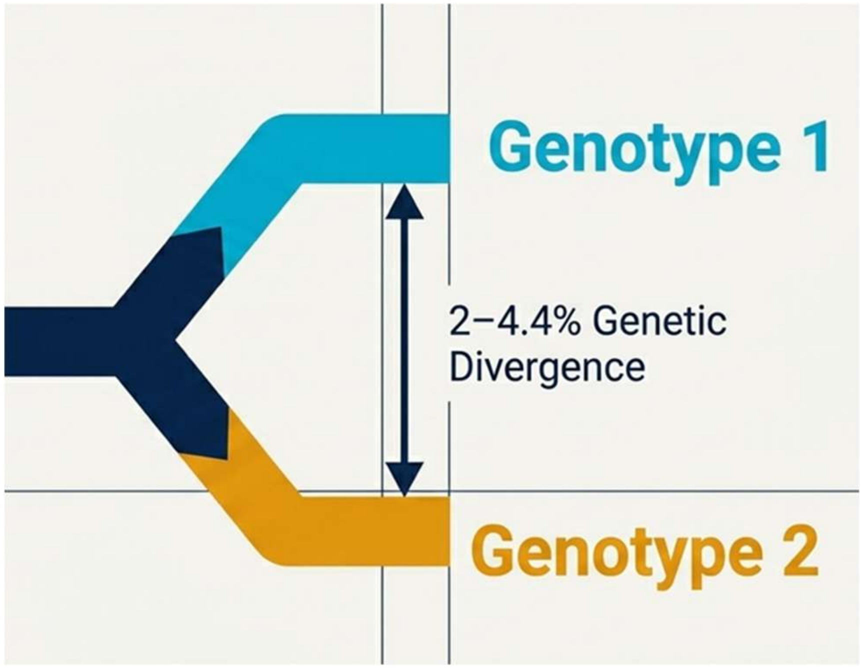
One Parasite, Two Genetic Identities

Phylogenetic analysis across all three tested loci demonstrated a consistent separation of *H. meleagridis* isolates into two distinct clusters.

The Genotype Designation

The genetic variability (2-4.4% difference) mirrors the divergence seen in related protozoa.

Therefore, these are not just clusters; they are two distinct **Genotypes**.



SUMMARY

The Flaw

ITS markers not suitable for tracking *H. meleagridis* due to intra-isolate heterogeneity.

A single cell produces conflicting genetic tags.

The Finding

The parasite exists as two reproducible genetic clusters: **Genotype 1 (widespread across Europe) and Genotype 2 (highly localized to France).**

The Fix

Combine *18S rRNA* and *rpb1* loci for accurate molecular subtyping

NEXT STEPS

Isolate and Purify

Isolate and purify Genotype 1 and Genotype 2

In Vivo Testing

Test the disease presentation and mortality associated with these two genotypes

Any questions

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